Polar Coordinates

$r = \sqrt{x^2 + y^2}$, $\Lambda^{\mu'}{}_{\mu} = \dfrac{\partial x^{\mu'}}{\partial x^{\mu}}$, $\det(\Lambda) \neq 0$ ← Jacobian of transformation.

$\phi = \tan^{-1}\left(\frac{y}{x}\right)$

$$V^{\mu'} = \Lambda^{\mu'}{}_{\mu} V^{\mu} \qquad e_{(\alpha')} = \Lambda^{\alpha}{}_{(\alpha')} e_{\alpha}$$

$V = V^{\alpha} e_{(\alpha)}$ ← basis vectors. ← elements

↑ basis independent vector

transforms opposite to vector comp. because V is coord. independent.

For Polar coords:

$$\begin{cases} e_{(r)} = \cos\varphi\, e_{(x)} + \sin\varphi\, e_{(y)} \\ e_{(\varphi)} = -r\sin\varphi\, e_{(x)} + r\cos\varphi\, e_{(y)} \end{cases}$$

$$\begin{cases} e_{(x)} = \cos\varphi\, e_{(r)} - \dfrac{\sin\varphi}{r} e_{(\varphi)} \\ e_{(y)} = \sin\varphi\, e_{(r)} + \dfrac{\cos\varphi}{r} e_{(\varphi)} \end{cases}$$

$$\frac{\partial e_{(r)}}{r} = 0$$? (not taking partials wrt to $e_{(x)}$, $e_{(y)}$)

$$\frac{\partial e_{(r)}}{\partial \varphi} = -\sin\varphi\, e_{(x)} + \cos\varphi\, e_{(y)} = \frac{e_{(\varphi)}}{r}$$

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$$\frac{\partial e_{(\varphi)}}{\partial \varphi} = -r\cos\varphi\, e_{(x)} + r\sin\varphi\, e_{(y)} = -r\, e_{(r)}$$

$$\frac{\partial e_{(\alpha)}}{\partial x^\beta} = \Gamma^{\mu}_{\alpha\beta}\, e_{(\mu)}$$

Christoffel symbol (connection coefficient).

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$$\Gamma^r_{rr} = \Gamma^\varphi_{rr} = 0 \;!$$

$$\left(\frac{\partial e_r}{\partial r}\right) e_r \qquad \left(\frac{\partial e_r}{\partial r}\right) e_\varphi$$

$$\Gamma^\varphi_{r\varphi} = \frac{1}{r}, \quad \Gamma^r_{r\varphi} = 0$$

$$\Gamma^\varphi_{\varphi r} = \frac{1}{r}, \quad \Gamma^r_{\varphi r} = 0$$

$$\Gamma^r_{\varphi\varphi} = -r, \quad \Gamma^\varphi_{\varphi\varphi} = 0$$

notice the symmetry!

$$\frac{\partial V}{\partial x^\beta} = \frac{\partial}{\partial x^\beta}\left(V^\alpha e_{(\alpha)}\right)$$

$$= \frac{\partial V^\alpha}{\partial x^\beta} e_{(\alpha)} + V^\alpha \frac{\partial e_{(\alpha)}}{\partial x^\beta}$$

$$= \frac{\partial V^\alpha}{\partial x^\beta} e_{(\alpha)} + V^\alpha \Gamma^{\gamma}_{\alpha\beta}\, e_\gamma$$

μ α

$$= \underbrace{\left(\frac{\partial V^\alpha}{\partial x^\beta} + V^\mu \Gamma^\alpha_{\mu\beta}\right) e_{(\alpha)}}$$

the covariant derivative.

V^α influence in the α-dir.

V^μ influence in the α-dir

Conventions

$$\frac{\partial V^\alpha}{\partial x^\beta} \text{ (partial)} = V^\alpha{}_{,\beta}$$

$$\underbrace{V^\alpha{}_{;\beta}}_{\text{covariant}} = V^\alpha{}_{,\beta} + V^\mu \Gamma^\alpha_{\mu\beta}$$

can we have an example?

α-component of $V_{;\beta}$

$$\vec{E} = (a, b) = a \cdot e_{(x)} + b \cdot e_{(y)}.$$ → constant electric field.

$$= a\left(\cos\varphi\, e_{(r)} - \frac{\sin\varphi}{r} e_{(\varphi)}\right) + b\left(\sin\varphi\, e_{(r)} + \frac{\cos\varphi}{r} e_{(\varphi)}\right)$$

$$= \underbrace{(a\cos\varphi + b\sin\varphi)}_{E^r} e_{(r)} + \underbrace{\frac{1}{r}(b\cos\varphi - a\sin\varphi)}_{E^\varphi} e_\varphi$$

$$E^r{}_{;r} = \frac{\partial E^r}{\partial r} + E^\alpha \Gamma^r_{\alpha r} = 0 + 0 = 0$$

$$E^r{}_{;\varphi} = \frac{\partial E^r}{\partial \varphi} + E^\alpha \Gamma^r_{\alpha\varphi}$$

$$= (-a\sin\varphi + b\cos\varphi) + \cancel{E^r \Gamma^r_{r\varphi}} + E^\varphi \Gamma^r_{\varphi\varphi}$$

$$= (-a\sin\varphi + b\cos\varphi) + \left(\frac{1}{r} b\cos\varphi - \frac{1}{r} a\sin\varphi\right)(-r)$$

$$= 0$$

this is a tensor

verified that this is a uniform Electric field, so $E^r{}_{;r}$ & $E^r{}_{;\varphi}$ should be 0!

$$E^\varphi{}_{;r} = \frac{\partial E^\varphi}{\partial r} + E^\alpha \Gamma^\varphi_{\alpha r} = \underbrace{\frac{\partial E^\varphi}{\partial r} + E^r \Gamma^\varphi_{rr} + E^\varphi \Gamma^\varphi_{\varphi r}}$$

$$= 0$$

I need a more physical intuition of $\Gamma^\alpha_{\mu\beta}$

a geometric one

invariant infinitesimal distance

$$ds^2 = dx^2 + dy^2 = (dr\cos\varphi - r\sin\varphi\, d\varphi)^2 + (dr\sin\varphi + r\cos\varphi\, d\varphi)^2$$

$$= dr^2 + r^2 d\varphi^2 = g_{\mu\nu}\, dx^\mu dx^\nu$$

metric tensor

$x = r\cos\varphi$

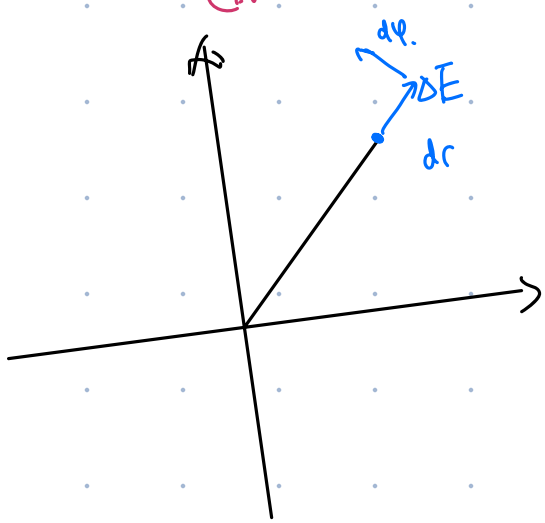
$dx = dr\cos\varphi - r\sin\varphi\, d\varphi$

(symmetric: $g_{\mu\nu} = g_{\nu\mu}$)

$$g_{\mu\nu} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$g_{\mu\nu} = \begin{pmatrix} 1 & 0 \\ 0 & r^2 \end{pmatrix}, \quad g^{\mu\nu} = \begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{r^2} \end{pmatrix}$$

Can I derive the metric for a sphere?



YES.

$$\boxed{V^\alpha{}_{;\beta}} = V^\alpha{}_{,\beta} + \Gamma^\alpha_{\rho\beta} V^\rho$$

tensor

$$\frac{\partial}{\partial x^\beta}\left(V^\alpha w_\alpha\right) \Rightarrow \text{this is a scalar?}$$

$$= \frac{\partial V^\alpha}{\partial x^\beta} w_\alpha + V^\alpha \frac{\partial w_\alpha}{\partial x^\beta}$$

$$= \cancel{V^\alpha{}_{,\beta}} w_\alpha + V^\alpha w_{\alpha,\beta}$$

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$\| $$

$$(V^\alpha w_\alpha)_{;\beta} = \boxed{V^\alpha{}_{;\beta}} w_\alpha + V^\alpha w_{\alpha;\beta}$$

$$\left(V^\alpha{}_{;\beta} + V^\mu \Gamma^\alpha_{\mu\beta}\right) w_\alpha + V^\alpha w_{\alpha;\beta}$$

$$V^\alpha w_{\alpha;\beta} = V^\alpha w_{\alpha,\beta} - V^\mu \Gamma^\alpha_{\mu\beta} w_\alpha$$

$$= V^\alpha w_{\alpha,\beta} - V^\alpha \Gamma^\gamma_{\alpha\beta} w_\gamma$$

$$\boxed{w_{\alpha;\beta} = w_{\alpha,\beta} - \Gamma^\gamma_{\alpha\beta} w_\gamma}$$

now we know how lower indices covariant derivatives work.

compare:

$$V^\alpha{}_{;\beta} = V^\alpha{}_{,\beta} + V^\mu \Gamma^\alpha_{\mu\beta}.$$

What about $\binom{2}{0}$ tensors?

$$\left(V^\alpha W^\beta\right)_{;\rho} = V^\alpha{}_{;\rho} W^\beta + V^\alpha W^\beta{}_{;\rho}$$

$$= \underline{V^\alpha{}_{,\rho} W^\beta + V^\alpha W^\beta{}_{,\rho}} \Leftrightarrow (V^\alpha W^\beta)_{,\rho}$$

$$+ V^\alpha \Gamma^\beta_{\sigma\rho} W^\sigma + \Gamma^\alpha_{\sigma\rho} V^\sigma W^\beta$$

$$\Downarrow$$

$$T^{\alpha\beta}{}_{;\rho} = T^{\alpha\beta}{}_{,\rho} + \Gamma^\alpha_{\sigma\rho} T^{\sigma\rho} + \Gamma^\beta_{\sigma\rho} T^{\alpha\sigma}$$